

Adithya Rao Kalathur

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🌐 LinkedIn | 🐙 GitHub | 📁 Portfolio | 🏆 Certificates

PROFILE

ML Engineer building **agentic AI systems, developer tooling, and retrieval architectures** for production-scale applications. Proficient in **Python, PyTorch, FastAPI, and Docker**. Seeking **ML engineering or AI backend roles**.

EDUCATION

Manipal School of Information Sciences (MSIS) <i>Master of Engineering in Computer Science and Engineering; CGPA: 7.96/10</i>	Manipal, India <i>2025 – 2027 (Expected)</i>
Manipal Institute of Technology (MIT) <i>Bachelor of Technology in Computer and Communication Engineering; CGPA: 7.06/10</i>	Manipal, India <i>2020 – 2024</i>

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, C++
AI Systems & LLMs: LangGraph, LangChain, FastMCP, FAISS, BM25, Ollama, Sentence Transformers, Cross-Encoder Reranking, RAG Pipelines
ML & Deep Learning: PyTorch, Scikit-Learn, TensorFlow, U-Net, Transfer Learning, Grad-CAM, Hugging Face, Optuna
NLP & Data Science: Natural Language Processing, Text Classification, Pandas, NumPy
Backend & Web: FastAPI, PostgreSQL, REST APIs, React.js, Node.js, Streamlit
Data & Analytics: Power BI, DAX, ETL Pipelines, SQL
Systems & Tools: Docker, Linux, Shell Scripting, Git/GitHub, OpenCV, API Integration

WORK EXPERIENCE

Full Stack Web Development Intern <i>Thaniya Technologies [Certificate]</i>	Jun 2023 – Jul 2023 <i>Mangalore, India</i>
- Built and deployed a full-stack movie review and recommendation platform using React.js, Node.js, and PostgreSQL; shipped to production on Render via cloud hosting [GitHub]. - Designed the relational database schema, implemented REST APIs for all CRUD operations, and integrated backend services with the React frontend.	
Freelance Web Developer <i>Startup Client Project</i>	Dec 2024 – Mar 2025 <i>Remote</i>
- Built and deployed a responsive portfolio website for a landscape design startup; managed client communication, iterated on designs based on feedback, and delivered within timeline. - Structured service and contact pages for desktop and mobile; sole point of contact for requirements, architecture, and deployment decisions.	

PROJECTS

CodeIntel-RAG — Agentic Codebase Intelligence Platform [GitHub] <i>RAG, Hybrid Retrieval, FAISS, BM25, LangGraph, FastMCP, FastAPI, Ollama, MCP Server, Security Analysis</i>	2025
- Built a self-hosted agentic RAG platform for natural-language querying over Python, Java, JavaScript, TypeScript, and C++ codebases; achieved Hit@5 of 1.00 and MRR of 0.845 on a 25-query benchmark against pallets/flask — 8.7% above the semantic baseline. - Designed a custom 5-signal hybrid retrieval pipeline combining BGE embeddings + FAISS, BM25Okapi, structural call-graph traversal, file-metadata filtering, and name-token concept matching. - Engineered a LangGraph ReAct agent and a spec-compliant FastMCP server sharing five tools (hybrid search, Bandit + regex security scanner with CWE/OWASP Top 10 mapping, multi-hop dependency analysis, call-graph retrieval, repo overview); MCP server supports external clients including Cursor and Continue.dev over SSE.	

- Deployed a FastAPI backend with 15+ endpoints, SSE streaming, query caching, and Docker Compose containerisation; frontend includes voice input (Whisper) and interactive Pyvis call graphs.

Deep Learning-Based SCC Grading from Histopathology Images [\[GitHub-1\]](#) [\[GitHub-2\]](#) 2025

Deep Learning, Transfer Learning, Computer Vision, Explainable AI (XAI), Grad-CAM, Streamlit, PyTorch, Optuna

- Benchmarked 5 CNN architectures across two tasks — adenocarcinoma vs SCC classification (InceptionV3, **96.5% accuracy, macro-F1 0.94**) and 3-class SCC grading (EfficientNet-B0, **89.15% accuracy, ROC-AUC 0.93**) on institutional histopathology datasets.
- Applied CLAHE and Reinhard stain normalisation for H&E stain consistency; used Optuna (150 trials) for automated hyperparameter search across learning rate, dropout, and regularisation.
- Deployed an interactive Streamlit web app with real-time Grad-CAM visualisations and LLM-generated prediction summaries, making model decisions interpretable for clinical users.
- Addressed class imbalance through class-weighted loss and extensive augmentation; applied two-phase transfer learning, early stopping, and learning rate scheduling to improve generalisation.

Night Vision Enhancement using Deep Learning [\[GitHub\]](#) 2025

U-Net, ResNet34, PyTorch, OpenCV, Computer Vision, Image-to-Image Translation

- Engineered two low-light image enhancement pipelines — a vanilla U-Net and a ResNet34 encoder-decoder U-Net — trained separately on the LOL dataset (485 pairs) and the ImageNet Night Vision dataset (40K+ pairs) using paired image-to-image translation.
- Achieved **PSNR of 18.84 dB** and **SSIM of 0.71** on the ImageNet Night Vision test set (5000 images) with the pretrained ResNet34 U-Net trained using L1 loss and ReduceLROnPlateau scheduling.
- Designed training pipelines with combined MSE + SSIM loss, ColorJitter augmentation, and residual learning; evaluated outputs quantitatively using PSNR and SSIM across multiple model variants.

FMCG Supply Chain Analytics Dashboard [\[GitHub\]](#) 2025

Power BI, DAX, SQL, ETL, Data Analytics

- Designed an end-to-end Power BI dashboard tracking supply chain KPIs — OTIF, LIFR, and VOFR — across 3 cities and 50+ product categories for Atliq Mart.
- Developed 20+ DAX measures and SQL-based ETL pipelines for root-cause analysis; identified a **15% fulfillment service gap** across key accounts that directly informed operational strategy.

PUBLICATIONS & RESEARCH

SQL Injection Detection: Comparative Study of SVM, RF and Hybrid Models [\[Publication\]](#) Oct 2025

SVM, Random Forest, BiLSTM, NLP, Hybrid Ensemble, Cybersecurity

- **First Author** — Published as a book chapter in *Connecting Intelligence: Trends in Computation and Data Communication*, Taylor & Francis / CRC Press (ISBN: 9781003773504); presented at ICCIDC 2025, Bali, Indonesia [\[Certificate\]](#)
- Achieved **97.84% accuracy, 98.09% precision**, and **97.57% recall** with a hybrid SVM-RF ensemble on 30,920 labeled SQL queries; a BiLSTM model further reached **98.75% accuracy**, validating sequence-based NLP for automated threat detection.

CERTIFICATIONS

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| • IBM Data Science Professional Certificate | Coursera, 2024 |
| • Google Data Analytics Professional Certificate | Coursera, 2024 |
| • Generative AI with Large Language Models | DeepLearning.AI, 2025 |
| • Google Cybersecurity Professional Certificate | Coursera, 2024 |

LANGUAGES

English: Professional proficiency

Kannada: Native

Tulu: Native

Hindi: Conversational